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Sub - Front End Technologies

Assignment

Ans(01)–

Redux is a state management library used in React to manage global application state in predictable way.

- Store → holds the state.
- action → object that describes what happened.
- Reducer → function that updates state based on action.
- Dispatch → sends action to reducer.

Benifits –

- Centralized State.
- Easy debugging.
- Predictable data flow.

Ans(02)–

Install Redux and React-Redux;

```
npm install redux react-redux
```

ex.

```
import {createStore} from "redux";
const initialstate = {
    count : 0
};
function counterReducer(state=initialState, action){
    switch (action.type){
        case "INCREMENT":
            return {count : state.count +1};
        case "DECREMENT":
            return {count : state.count -1};
        default:
            return state;
    }
}
const store = createStore (counterReducer);
export default store;
```

ex.

```
import React from "react";
import ReactDOM from "react-dom/client";
import {provider} from "react-redux";
import App from "./app";
import store from "./store";

const root = ReactDOM.createRoot(document.
                                 getElementById("root"));

root.render(
    <provider store={store}>
        <App/>
    </provider>
);
```

Ans(03)-

ex.

```
function App() {
    const count = useSelector((state) => state.count);
    const dispatch = useDispatch();
    return(
        <div>
            <h2> Count: {count} </h2>
            <button onclick={() => dispatch(
                        {type: "INCREMENT"})} >
                Increment
            </button>
            <button onClick={() => dispatch(
                        {type: "DECREMENT"})} >

                Decrement
            </button>
        </div>
    );
}
export default App;
```